Exercise for Sect. 2.1

Exercise 2.1.1

Show that the space $(\mathbb{T}, \mathcal{B}_{\mathbb{T}}, m_{\mathbb{T}})$ is isomorphic as a measure space to $(\mathbb{T}^2, \mathcal{B}_{\mathbb{T}^2}, m_{\mathbb{T}^2})$.

Proof: For any $x \in \mathbb{T}$, express x in an infinite series $(x_i)_{i\in\mathbb{N}}$ by x_i is the i-th binary place of x. Let $D = \left\{\frac{a}{2^k} : a, k \in \mathbb{N} \text{ and } a < 2^k\right\}$ denote the set of dyadic rationals and D is measure zero since D is countable. Then, define $A^2 \subset \mathbb{T}^2$ by $A^2 = T^2 \setminus (\mathbb{T} \times D \cup D \times \mathbb{T})$. A^2 is measure 1 since $D \times \mathbb{T}$ and $\mathbb{T} \times D$ are measure zero. In this construction, A^2 is the set only containing the points have infinitely many 0 and 1 digits. Define $f : 0.x_1x_2x_3\cdots \mapsto (0.x_1x_3x_5\cdots, 0.x_2x_4x_6\cdots)$ and $A = f^{-1}(A^2)$. A is measurable and measure 1 since $A = \mathbb{T} \setminus (D \cup D_{\text{odd}} \cup D_{\text{even}})$ where the two sets stand for the set of points contiains fintely many 0 or 1 in odd or even digits.

To prove f is bimeasurable, define the cylender set $I_{x_1,x_2,\ldots,x_n} = \{y \in A : y_i = x_i \text{ for } i \le n\}$ and collection of the intervals $\mathcal{S} := \left\{I_{x_1,\ldots,x_n} : x_i \in \{0,1\} \text{ and } n \in \mathbb{N}\right\}$ which is a π-system on A that $\sigma(\mathcal{S}) = \mathcal{B}_{\mathbb{T}}$. Similarly, $\mathcal{S}^2 := \left\{R_{(x_1,\ldots,x_n),(y_1,\ldots,y_m)}\right\}$ where $R_{(x_1,\ldots,x_n),(y_1,\ldots,y_m)} = \left\{(z,w) \in A^2 : z_i = x_i, w_j = y_j \text{ for } i \le n \text{ and } j \le m\right\}$ and $\sigma(\mathcal{S}^2) = \mathcal{B}_{\mathbb{T}^2}$. Then, we want to show f and f^{-1} maps the cylinder sets into measurable sets.

Given an $I_{x_1,x_2,\ldots,x_{2k}}$, $f\left(I_{\ldots,x_{2k}}\right) = R_{(x_1,\ldots,x_{2k-1}),(x_2,\ldots,x_{2k})}$, and given $I_{x_1,x_2,\ldots,x_{2k+1}}$ get $f(I) = R_{(\ldots,x_{2k+1}),(\ldots,x_2k)}$. These two situation is trivially measurable.

In contrast, for given $R_{(\ldots,x_n),(\ldots,y_m)}$, we can divide it into finite disjoint union of cylinders $R_{(\ldots,x_k),(\ldots,y_k)}$ where $k \le \max(n,m)$. Then the preimage is easy to calculate, $f^{-1}\left(R_{(\ldots,x_k),(\ldots,y_k)}\right) = I_{x_1,y_1,\ldots,x_k,y_k}$ measurable.

Last is to check if f preserves measure. $m_{\mathbb{T}}\left(I_{\ldots,x_k}\right) = 2^{-k}$ and $m_{\mathbb{T}^2}\left(R_{(\ldots,x_n),(\ldots,y_m)}\right) = 2^{-n} \times 2^{-m} = 2^{-nm}$. From the result of mapping above, f and f^{-1} preserves measure on cylinder sets and thus preserves on all measurable sets. Thus, $(\mathbb{T}, \mathcal{B}_{\mathbb{T}}, m_{\mathbb{T}})$ is isomorphic as a measure space to $(\mathbb{T}^2, \mathcal{B}_{\mathbb{T}^2}, m_{\mathbb{T}^2})$. $\square$

Exercise **2.1.2**

Show that the measure-preserving system $(\mathbb{T}, \mathcal{B}_{\mathbb{T}}, m_{\mathbb{T}}, T_4)$, where $T_4(x) = 4x (\mathrm{mod}\ 1)$, is measurably isomorphic to the product system $(\mathbb{T}^2, \mathcal{B}_{\mathbb{T}^2}, m_{\mathbb{T}^2}, T_2 \times T_2)$.

Proof: By the construction above, this is trivial by treating T_k as moving all digit leftward k times. □

Exercise **2.1.3**

For a map $T : X \to X$ and sets $A, B \subseteq X$, prove the following.

- $\chi_A(T(x)) = \chi_{T^{-1}(A)}(x)$;
- $T^{-1}(A \cap B) = T^{-1}(A) \cap T^{-1}(B)$;
- $T^{-1}(A \cup B) = T^{-1}(A) \cup T^{-1}(B)$;
- $T^{-1}(A \Delta B) = T^{-1}(A) \Delta T^{-1}(B)$.

Which of thexe properties also hold with the pre-image under T^{-1} replaced by the forward image under T?

Proof: The four are easy and after replacing, only union still holds. □

Exercise **2.1.4**

Let X be a compact abelian group and let $T : X \to X$ be a countinuous homomorphism. Does T still preserve the Haar measure m_X on X?

Proof: Let T be the constant map from X to identity of X, T satisfies the assumption and may not preserve measure if X contains more than one elements. $\square$

Exercise **2.1.5**

1. Find a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with a non-trivial factor map $\varphi : X \to X$.
2. Find an invertible measure-preserving system $(X, \mathcal{B}, \mu, T)$ with a non-trivial factor map $\varphi : X \to X$.

Proof:

1. $X = \{0, 1\}^{\mathbb{N}}$ be the sample space of a infinite series of outcome of Bernoulli trials, and $\mathcal{B}$ is the σ-algebra generate by all cylinder sets. μ is the Bernulli measure with probability $\frac{1}{2}$ and $\frac{1}{2}$. Define T by $(Tx)_n = x_{n+1}$ for $x \in X$ which is the left shift map on X. Then, claim that $\varphi(x_n) = x_n + x_{n+1} \bmod 2$ is a non-trivial factor map on X to itself.

   First, φ is not an isomorphism since it maps two elements (one solution and its comjugate) to one element, and φ is surjective since for $y \in X, y = y + 0 \bmod 2$. Then, $(\varphi \circ T(x))_n = \varphi\big((Tx)_n\big) = \varphi(x_{n+1}) = x_{n+1} + x_{n+2} \bmod 2 = (\varphi(x))_{n+1} = T\big((\varphi(x))_n\big) = (T \circ \varphi(x))_n$.

   Last, for C is an cylinder set given by $\left\{x \in X : x_{n_1} = c_1, \ldots, x_{n_k} = c_k\right\}$. If n_i and n_j are not next to each other, the choice of each one won't affect to another one, so we can divides into not adjacency parts and multiply them after computing each part. Thus, we only focus on the situation that $n_j = n_i + 1$, $n_l = n_j + 1$ and so on. If $k = 1$, $\mu\big(\varphi^{-1}(C)\big) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2} = \mu(C)$(11 and 00 or 01 and 10.) If $k = 2$, $\mu\big(\varphi^{-1}(C)\big) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot 2 = \frac{1}{4} = \mu(C)$ (the whole process actually depends on the first one.) In conclusion, $\mu\big(\varphi^{-1}(C)\big) = \mu(C)$ for all cylinder sets and thus φ is measure-preserving on $\mathcal{B}$.
2. Just modify X into $\{0, 1\}^{\mathbb{Z}}$, the other settings are still the same.

$\square$

Exercise **2.1.6**

Prove that the circle rotation $R_\alpha : \mathbb{T} \to \mathbb{T}, R_\alpha(t) = t + a_\alpha (\text{mod } 1)$ is not measurably isomorphic to the circle-doubling map $T_2 : \mathbb{T} \to \mathbb{T}, T_2(t) = 2t(\text{mod } 1)$.

Proof: Assume there is an isomorphic measurable function $\varphi : \mathbb{T} \to \mathbb{T}$ such that $R_\alpha \circ \varphi = \varphi \circ T_2$ almost everywhere. Then, for almost every $x \in \left(0, \frac{1}{2}\right)$ satisfies the equation,

$$\varphi(x) + \alpha \text{ mod } 1 = \varphi(2x)$$

$$(\varphi(2x) - \varphi(x)) \text{ mod } 1 = \alpha$$

$$\Rightarrow \begin{cases} \varphi(2x) - \varphi(a) = \alpha \\ \varphi(2x) - \varphi(b) = \alpha - 1' \end{cases}$$

where one of a, b is x and another is $x + \frac{1}{2}$. Minus that two equations can get $\varphi(b) - \varphi(a) = 1$; however, it is impossible since range of φ is $[0, 1)$. Thus, the two map are not measurable isomorphic. $\square$

Exercise **2.1.7**

Let $\mathfrak{X} = (X, \mathcal{B}, \mu, T)$ be any measure-preserving system. A sub-σ-algebra $\mathcal{A} \subseteq \mathcal{B}_X$ with $T^{-1}\mathcal{A} = \mathcal{A}$ modulo μ is called a *T-invarient sub-σ-algebra*. Show that the system $\tilde{\mathfrak{X}} = \left(\tilde{X}, \tilde{\mathcal{B}}, \tilde{\mu}, \tilde{T}\right)$ defined by

- $\tilde{X} = \left\{x \in X^{\mathbb{Z}} : x_{k+1} = T\left(x_k\right) \text{ for all } k \in \mathbb{Z}\right\}$;
- $\left(\tilde{T}(x)\right)_k = x_{k+1}$ for all $k \in \mathbb{Z}$ and $x \in \tilde{X}$;
- $\tilde{\mu}\left(\left\{x \in \tilde{X} : x_0 \in A\right\}\right) = \mu(A)$ for any $A \in \mathcal{B}$, and $\tilde{\mu}$ is invariant under $\tilde{T}$;
- $\tilde{\mathcal{B}}$ is the smallest $\tilde{T}$-invariant σ-algebra for which the map $\pi : x \mapsto x_0$ from $\tilde{X}$ to X is measurable;

is an invertible measure-preserving system, and that the map $\pi : x \mapsto x_0$ is a factor map. The system $\tilde{\mathfrak{X}}$ is called the invertible extension of $\mathfrak{X}$.

Proof:

- Invertible: $\left(\tilde{T}^{-1}(x)\right)_k = x_{k-1}$ is clearly the inverse map, and $\tilde{T}^{-1}$ is also measurable.
- Measure-preserving: $\tilde{\mu}$ is invarant under $\tilde{T}$ implies the measure-preserving condition.
- Factor map: for any $x \in \tilde{X}$, $\pi \circ \tilde{T}(x) = x_1 = T\left(x_0\right) = T(\pi(x))$.

$\square$

Exercise **2.1.8**

Show that the invertible extension $\tilde{\mathfrak{X}}$ of a measure-preserving system $\mathfrak{X}$ constructed above has the following universal property. For any extension

$$\varphi : (Y, \mathcal{B}_Y, \nu, S) \to (X, \mathcal{B}_X, \mu, T)$$

for which S is invertible, there exists a unique map

$$\tilde{\varphi} : (Y, \mathcal{B}_Y, \nu, S) \to \left(\tilde{X}, \tilde{\mathcal{B}}, \tilde{\mu}, \tilde{T}\right)$$

for which $\varphi = \pi \circ \tilde{\varphi}$.

Proof: Let $\tilde{\varphi}(y_0) = \left(\ldots, \varphi\left(S^{-2}y_0\right), \varphi\left(S^{-1}y_0\right), \varphi\left(y_0\right), \varphi\left(Sy_0\right), \varphi\left(S^2y_0\right), \ldots\right)$, and $\tilde{\varphi}$ satisfies $\varphi = \pi \circ \tilde{\varphi}$. Also, $\tilde{\varphi} \circ S = \tilde{T} \circ \tilde{\varphi}$ and $\tilde{\varphi} \circ S^n = \tilde{T}^n \circ \tilde{\varphi}$ for all $n \in \mathbb{Z}$.

If there is another map $\tilde{\varphi}'$ such that $\varphi = \pi \circ \tilde{\varphi}'$, for any $y \in Y$, $\pi \circ \tilde{\varphi}'(y) = \varphi(y) = \pi \circ \tilde{\varphi}(y)$. Using the projection map and the property above,

$$(\tilde{\varphi}'(y))_k = \pi\left(\tilde{T}^k \circ \tilde{\varphi}(y)\right) = \pi(\tilde{\varphi} \circ S^n(y)) = (\tilde{\varphi}(y))_k.$$

Thus, the map is unique. □

Exercise **2.1.9**

1. Show that the invertible extension of the circle-doubling map T_2,

$$X_2 = \{x \in \mathbb{T}^{\mathbb{Z}} : x_{k+1} = T_2 x_k \text{ for all } k \in \mathbb{Z}\},$$

is a compact abelian group with respect to the coordinate-wise addition defined by $(x+y)_k = x_k + y_k$ for all $k \in \mathbb{Z}$, and the topology inherited from the product topology on $\mathbb{T}^{\mathbb{Z}}$.

2. Show that the diagonal embedding $\delta(r) = (r, r)$ embeds $\mathbb{Z}\left[\frac{1}{2}\right]$ as a descrete subgroup of $\mathbb{R} \times \mathbb{Q}_2$, and that $X_2 \cong \mathbb{R} \times \mathbb{Q}_2 / \delta\left(\mathbb{Z}\left[\frac{1}{2}\right]\right) \cong \mathbb{R} \times \mathbb{Z}_2 / \delta(\mathbb{Z})$ as compact abelian groups. In particular, the map $\tilde{T}_2$ (which may be thought of as the left shift on X_2, or as the map that doubles in each coordinate) is conjugate to the map

$$(s, r) + \delta\left(\mathbb{Z}\left[\frac{1}{2}\right]\right) \mapsto (2s, 2r) + \delta\left(\mathbb{Z}\left[\frac{1}{2}\right]\right)$$

on $\mathbb{R} \times \mathbb{Q}_2 / \delta\left(\mathbb{Z}\left[\frac{1}{2}\right]\right)$. The group X_2 constructed in this exercise is a simple example of a *solenoid*.

Proof:

1. If $x, y \in X_2$, $(x+y)_k = x_k + y_k = T_2 x_{k-1} + T_2 y_{k-1} = T_2\left(x_{k-1} + y_{k-1}\right) = T_2(x+y)_{k-1}$. The zero sequence 0 is the identity of X_2 and $(x+y)_k = \left(y + x_k\right)$ for all k implies X_2 is an abelian group.

   Since a sequence in a product topology converges if and only if it converges pointwise, the compactness is true by the compactness of each $\mathbb{T}$.

2.

□

Exercise for Sect. 2.2

Exercise **2.2.1**

Prove the following version of Poincare recurrence with a weaker hypothesis (finite additivity in place of countable additivity for the measure) and with a stronger conclusion (a bound on the return time). Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system with μ only assumed to be a finitely additive measure, and let $A \in \mathcal{B}$ have $\mu(A) > 0$. Show that there is some positive $n \leq \frac{1}{\mu(A)}$ for which $\mu(A \cap T^{-n}A) > 0$.

Proof: Suppose $n \in \mathbb{N}$ such that $\mu\big(A \cap T^{-k}(A)\big) = 0$ for all $k \leq n$, then $\mu\big(T^{-j}(A) \cap T^{-k}(A)\big) = 0$ for all $j, k \leq n$ by measure-preserving of T. Therefore,

$$\mu(X) \geq \mu\left(\bigcup_{k=0}^{n} T^{-k}(A)\right) = \sum_{k=1}^{n} T^{-k}(A) = n \cdot \mu(A).$$

The condition of $(X, \mathcal{B}, \mu)$ is a probability space implies the upper bound of n is $\frac{1}{\mu(A)}$.

□

Exercise 2.2.2

1. If $A \subseteq \mathbb{N}$ has positive density, meaning that

$$d(A) = \lim_{k\to\infty} \frac{1}{k}|A \cap [1,k]|$$

exists and is positive, prove that there is some $n \geq 1$ with $\overline{d}(A \cap (A - n)) > 0$ (here $A - n = \{a - n : a \in A\}$), where

$$\overline{d}(B) = \limsup_{k\to\infty} \frac{1}{k}|B \cap [1,k]|.$$

2. Can you prove this starting with ther weaker assumption that the upper density $\overline{d}(A)$ is positive, and reaching the same conclusion?

Proof:

1. First claim that $\overline{d}$ is a finite additive measure on $\mathbb{P}(\mathbb{N})$.
   - well-defined: for any $B \in \mathbb{P}(\mathbb{N})$, let $d_k = \frac{1}{k}|B \cap [1,k]|$, then $\overline{d}(B) = \limsup d_k$ uniquely exists since d_k is a real sequence.
   - on $[0,1]$: $\overline{d}(\varnothing) = \limsup \frac{1}{k} \cdot 0 = 0$, $d_k \in [0,1]$ for all k implies $\overline{d} : \mathbb{P}(\mathbb{B}) \to [0,1]$.
   - finite additive: if A, B are disjoint, $|(A \cup B) \cap [1,k]| = |A \cap [1,k]| + |B \cap [1,k]|$ for all k implies the finite additivity.

   Then, $\left(\mathbb{N}, \mathbb{P}(\mathbb{N}), \overline{d}\right)$ is a probility space and claim $T(x) = x + 1$ is a measure-preserving map. For all $A \in \mathbb{P}(\mathbb{N})$,

$$\begin{aligned}
\overline{d}\left(T^{-1}(A)\right) &= \limsup_{k\to\infty} \frac{1}{k}|(A-1) \cap [1,k]| \\
&\leq \limsup_{k\to\infty} \frac{1}{k}(|A \cap [1,k]| + 1) \\
&\leq \limsup_{k\to\infty} \frac{1}{k}|A \cap [1,k]| + \limsup_{k\to\infty} \frac{1}{k} \\
&= \limsup_{k\to\infty} \frac{1}{k}|A \cap [1,k]| = \overline{d}(A),
\end{aligned}$$

   and

$$
\begin{aligned}
\overline{d}\big(T^{-1}(A)\big) &= \limsup_{k\to\infty} \frac{1}{k} |(A-1) \cap [1,k]| \\
&\geq \limsup_{k\to\infty} \frac{1}{k} (|A \cap [1,k]| - 1) \\
&\geq \limsup_{k\to\infty} \frac{1}{k} |A \cap [1,k]| - \limsup_{k\to\infty} \frac{1}{k} \\
&= \overline{d}(A).
\end{aligned}
$$

Since A has positive density, $\overline{d}(A) > 0$, then there is an $n \in \mathbb{N}$ satisfies that $\overline{d}(A \cap T^{-n}(A)) > 0$ by the result in 2.2.1.

2. Trivially proved by the arguments above.

$\square$

Exercise **2.2.3**

1. Let (X, d) be a compact metric space and let $T : X \to X$ be a continuous map. Suppose that μ is a T-invariant probability measure defined on the Borel subsets of X. Prove that for μ-almost every $x \in X$ there is a sequence $n_k \to \infty$ with $T^{n_k}(x) \to x$ as $k \to \infty$.
2. Prove that the same conclusion holds under the assumption that X is a metric space, $T : X \to X$ is Borel measurable, and μ is a T-invariant probability measure.

Proof:

1. For all $k \in \mathbb{N}$, let $\cup_{x \in X} D\left(x, \frac{1}{k}\right)$ be the union of open disks on X. Since X is compact, there is finite set I_k such that $\cup_{x \in I} D\left(x, \frac{1}{k}\right)$ cover X. Then, suppose $F_k \subset I_k$ such that $\mu\left(D\left(x, \frac{1}{k}\right)\right) > 0$ for all $x \in I_k \setminus F_k$ and is 0 for all $x \in F_k$. By Poincare's theorem, for $x \in I_k \setminus F_k$ and a.e. $y \in D\left(x, \frac{1}{k}\right)$, $T^n(y) \in D\left(x, \frac{1}{k}\right)$ i.o. Thus, let $G_{x,k} = \left\{y \in D\left(x, \frac{1}{k}\right) : T^n(y) \text{ not back to } D\left(x, \frac{1}{k} \text{ i.o.}\right)\right\}$ and it is measure zero. Union of the sets $D\left(x, \frac{1}{k}\right)$ for $x \in F_k$ and $G_{x,k}$, called E_k, is measure zero since it is union of finite measure zero sets. Last, the union of E_k for all $k \in \mathbb{N}$ is still measure zero and for all $x \in X \setminus \left(\cup E_k\right)$ satisfies $T^{n_k} x \to x$ as $k \to \infty$ by choosing indices from each step.
2. Suppose $S = \operatorname{supp}(\mu) := \{x \in X : \mu(D(x, r)) > 0 \forall r > 0\}$.

   Claim that $\mu(S) = 1$. Observe on the set $S^c = X \setminus S$, $x \in S^c$ implies there is some $r_x > 0$ such that $\mu\left(D\left(x, r_x\right)\right) > 0$. Also, the $D\left(x, r_x\right)$ is the open ball contains in S^c, which implies S^c is open. Therefore, $\mu(S^c) = \mu\left(\cup_{x \in S^c} D\left(x, r_x\right)\right)$ by open. For any countable index set $\mathbb{I} \subset S^c$, we have

   $$\mu\left(\cup_{x \in \mathbb{I}} D\left(x, r_x\right)\right) \leq \sum_{x \in \mathbb{I}} \mu\left(D\left(x, r_x\right)\right) = \sum_{x \in \mathbb{I}} 0 = 0.$$

   By the definition of uncountable summation,

   $$\mu(S^c) \leq \sum_{x \in S^c} \mu\left(D\left(x, r_x\right)\right) = \sup_{\mathbb{I}} \sum_{x \in \mathbb{I}} \mu\left(D\left(x, r_x\right)\right) = 0.$$

Thus, $\mu(S) = 1$.

Then, for any $r > 0$.

Claim that S is cover by at most countable open ball with radius r Suppose R is a collection of points in S such that $x, y \in R$, $d(x, y) > r$. Then, the set $C = \cup_{x \in R} D\left(x, \frac{r}{2}\right)$ is a union of disjoint sets in R. By the assumption that $D\left(x, \frac{r}{2}\right)$ has positive measure for any $x \in S$, $\frac{r}{2} > 0$, we have $\mu(C) = \sum_{x \in R} \mu\left(D\left(x, \frac{r}{2}\right)\right)$. If R is uncountable, the summation of uncountable positive real numbers goes to infinity, contradicts to the condition of probability space. Thus, R is at most countable and S is cover by $\cup_{x \in \mathbb{R}} D(x, r)$ by finding the biggest R satisfies the assumption.

Let $r = \frac{1}{k}$ for each k, and by the similar prove of 1., we have for almost every $x \in S$ returns to its neighborhood infinitely often with distance at most $2r$. Then take intersection of each k, we have for almost every x, $T^{n_k} x$ converges to x.

$\square$